WisdomBench-Embodied: A Longitudinal Benchmark for Measuring Learning Ability in Physical Agents

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Abstract

Existing embodied AI benchmarks—RLBench, BEHAVIOR-1K, RoboFail—evaluate agents on first-attempt success rate, measuring *what a robot can do* (intelligence). None measure *how well a robot learns from its operational experience* (wisdom). We introduce **WisdomBench-Embodied (WB-E)**, a longitudinal evaluation protocol that overlays a 5-round, multi-trial assessment on top of any existing manipulation benchmark. WB-E produces two orthogonal scores: Intelligence I_E (Round-1 success) and Wisdom W_E (Round-5 $-$ Round-1 improvement), along with three diagnostic axiom measurements: plasticity $\hat{\Phi}_E$, immunity $\hat{\Psi}_E$, and homeostasis $\hat{H}_E$. We apply WB-E to 35 tasks from RLBench, evaluating 12 agent configurations (4 architectures $\times$ 3 model capacities). Key findings: (1) I_E and W_E are decorrelated ($\rho = +0.09$, n.s.), revealing that benchmarks reporting only I_E miss an entire dimension of capability; (2) the axiom profile $(\hat{\Phi}_E, \hat{\Psi}_E, \hat{H}_E)$ predicts W_E with $R^2 = 0.81$; (3) two agents with identical I_E can differ by $10\times$ in W_E, depending on architecture. WB-E is open-source, requires no additional simulation infrastructure, and can be applied retroactively to any existing benchmark dataset with multi-trial recordings.

1 Introduction

Consider the following scenario: two VLA models—Model A (3B parameters, $I_E = 55\%$) and Model B (55B parameters, $I_E = 85\%$)—are evaluated on a shelf-organization task. Model B succeeds on 30% more tasks on the first attempt. By any existing benchmark, Model B is strictly superior.

But observe what happens when both models fail and attempt the same task again. Model A, equipped with a Cognitive Immunity architecture, succeeds on 12% more tasks by round 5 ($W_E = +12\%$). Model B, with no memory mechanism, shows zero improvement ($W_E = 0\%$).

Which model would you deploy in a warehouse that operates 24/7 for years?

The missing dimension. All major embodied benchmarks measure a single temporal snapshot: first-attempt success rate. This captures *intelligence*—the capacity to perform correctly under known conditions. But it misses *wisdom*—the capacity to improve under operational pressure, learn from errors, and transfer lessons across tasks [8, 9].

WisdomBench-Embodied. We propose WB-E, a benchmark protocol (not a new task suite) that adds the temporal dimension to any existing evaluation. It produces three outputs:

1. **IW Scatter Plot**: A 2D visualization revealing whether I_E and W_E are correlated or decorrelated for a given agent population.
2. **Axiom Profile**: Three diagnostic values $(\hat{\Phi}_E, \hat{\Psi}_E, \hat{H}_E)$ that *explain* the observed W_E.
3. **Learning Curve**: Round-by-round success rate, revealing the agent's learning dynamics.

Why this matters. As embodied agents move from lab demonstrations to long-term deployment [2], their ability to learn in the field becomes as important as their initial capability. WB-E provides the measurement tool the community currently lacks.

2 Related Work

Embodied benchmarks. RLBench [3] provides 100+ tasks for evaluating manipulation policies, reporting per-task success rate. BEHAVIOR-1K [4] offers 1,000 daily activities with long-horizon evaluation. RoboFail [6] provides 130 failure demonstrations. All measure single-trial performance.

Multi-trial evaluation. Some work reports "multi-trial" results (e.g., 10 trials per task), but these are independent identically-distributed samples—not sequential rounds where the agent could learn. WB-E is the first protocol to evaluate *sequential* improvement.

Wisdom measurement. WisdomBench [10] introduced the Wisdom Quotient for language agents. We extend this concept to physical agents, adapting the measurement protocol to trajectory-level observations.

3 The WB-E Protocol

3.1 Overview

WB-E augments any existing benchmark $\mathcal{B} = \{t_1, \ldots, t_K\}$ with a longitudinal layer:

Definition 1 (WB-E Evaluation)**.** *For an agent M with architecture $\mathcal{A}$, evaluate each task $t \in \mathcal{B}$ over $R = 5$ sequential rounds. After each failure in round r, provide the agent with a* failure observation *$o_r = (\textit{RGB}, \textit{depth}, \textit{proprioception}, \textit{trajectory})$. Record success $s_r(t) \in \{0, 1\}$ and trajectory $\tau_r(t) \in \mathbb{R}^{T \times d}$ for each round.*

3.2 Core Metrics

Definition 2 (Embodied Intelligence)**.**

$$I_E(M) = \frac{1}{K} \sum_{t=1}^{K} s_1(t) \tag{1}$$

The fraction of tasks the agent succeeds on its first attempt.

Definition 3 (Embodied Wisdom Quotient)**.**

$$\mathrm{EWQ}(M) = \frac{1}{K} \sum_{t=1}^{K} [s_R(t) - s_1(t)] \tag{2}$$

The net improvement between the first and final round, averaged over tasks. $\mathrm{EWQ} > 0$*: the agent learns.* $\mathrm{EWQ} = 0$*: no learning.* $\mathrm{EWQ} < 0$*: regression.*

Definition 4 (Per-Round Learning Rate)**.**

$$\lambda_r(M) = \frac{1}{K} \sum_{t=1}^{K} [s_r(t) - s_{r-1}(t)], \quad r = 2, \ldots, R \tag{3}$$

The incremental improvement in each round, capturing the learning dynamics.

3.3 Diagnostic Axiom Measurements

WB-E also computes three architectural diagnostics:

Plasticity $\hat{\Phi}_E$: Are the agent's trajectories changing across rounds?

$$\hat{\Phi}_E = \frac{1}{K} \sum_{t=1}^{K} \frac{\mathrm{DTW}(\tau_1(t), \tau_R(t))}{\mathrm{DTW}_{\max}} \tag{4}$$

Immunity $\hat{\Psi}_E$: Does failure on one task help on related tasks?

$$\hat{\Psi}_E = \frac{1}{|\mathcal{P}|} \sum_{(t_a, t_b) \in \mathcal{P}} \left[P(s^{t_b} | \text{fail}^{t_a}) - P(s^{t_b}) \right] \tag{5}$$

where $\mathcal{P}$ is the set of structurally related task pairs.

Homeostasis $\hat{H}_E$: Is the wisdom robust to perturbations?

$$\hat{H}_E = 1 - \frac{|\text{EWQ}(\mathcal{T}') - \text{EWQ}(\mathcal{T})|}{\text{EWQ}(\mathcal{T})} \tag{6}$$

where $\mathcal{T}'$ includes spatial (±5cm), visual (×0.5–2.0 lighting), and physical (±30% friction) perturbations.

3.4 Output Artifacts

WB-E produces the following standardized outputs:

Table 1: WB-E output specification.

Artifact	**Description**
`iw_scatter.pdf`	I_E vs. EWQ scatter with regression line and Spearman ρ
`learning_curve.pdf`	Round-by-round success rate per architecture
`axiom_profile.pdf`	Bar chart of $(\hat{\Phi}_E, \hat{\Psi}_E, \hat{H}_E)$
`results.json`	Raw per-task, per-round success/failure data
`report.md`	Auto-generated analysis report

4 Experimental Validation

4.1 Setup

We apply WB-E to 35 RLBench tasks organized in 7 categories (grasp, place, stack, pour, tool use, drawer, assembly), with 4 architectures × 3 model capacities × 5 rounds × 3 seeds = 6,300 trials.

4.2 Result 1: The Missing Dimension

Table 2 demonstrates the central thesis: a benchmark that reports only I_E would rank the 55B Vanilla model first and the 3B Cognitive Immunity model last. WB-E reveals that the ranking inverts on the wisdom dimension.

4.3 Result 2: Architecture as the Explanatory Variable

We confirm that the axiom profile $(\hat{\Phi}_E, \hat{\Psi}_E, \hat{H}_E)$ explains 81% of the variance in EWQ via power-law regression:

$$\log \text{EWQ} = 0.38 \log \hat{\Phi}_E + 0.29 \log \hat{\Psi}_E + 0.25 \log \hat{H}_E + C, \quad R^2 = 0.81 \tag{7}$$

This validates that WB-E's diagnostic axiom measurements are not just descriptive—they are *predictive*.

4.4 Result 3: Leaderboard Redesign

We propose a dual-axis leaderboard for embodied AI:

The composite score $S = 0.5 \cdot I_E + 0.5 \cdot \text{EWQ}$ (normalized) produces a fundamentally different ranking than I_E alone. Notably, the 55B Vanilla model—the "top" model by intelligence—drops to rank 10.

Table 2: WB-E reveals the hidden dimension. These two agents look identical on standard benchmarks but differ drastically in learning ability.

Agent	I_E	EWQ	RLBench Rank	**WB-E Verdict**
55B + Vanilla	84.7%	+0.3%	#1	Cannot learn
3B + Cog. Imm.	55.2%	+12.1%	#12	Fastest learner

Table 3: WB-E dual-axis leaderboard (sorted by composite score).

Rank	**Agent**	$I_E\%$	EWQ%	$\hat{\Phi}_E$	$\hat{\Psi}_E$	$\hat{H}_E$
1	55B + Cog. Imm.	86.3	+10.4	0.49	0.38	0.44
2	12B + Cog. Imm.	71.0	+11.8	0.49	0.38	0.44
3	55B + Reflexion	85.9	+6.8	0.52	0.28	0.13
4	3B + Cog. Imm.	55.2	+12.1	0.49	0.38	0.44
5	12B + Reflexion	70.5	+7.1	0.52	0.28	0.13
10	55B + Vanilla	84.7	+0.3	0.02	0.00	0.01

5 Usage Guide

5.1 Applying WB-E to Your Benchmark

WB-E requires no new simulation infrastructure. To apply it to an existing benchmark:

1. **Select tasks**: Choose $K \geq 20$ tasks from your benchmark.
2. **Group tasks**: Organize into ≥ 4 categories of structurally related tasks (for $\hat{\Psi}_E$).
3. **Run evaluation**: For each task, run $R = 5$ sequential rounds. After each failure, provide the failure observation to the agent.
4. **Run perturbation suite**: Re-run the 5-round evaluation under spatial, visual, and physical perturbations (for $\hat{H}_E$).
5. **Compute metrics**: Use our open-source `wisdombench-e` library.
6. **Generate report**: Automatically produces IW scatter, learning curves, axiom profiles, and dual-axis leaderboard.

5.2 Minimum Requirements

- $K \geq 20$ tasks (more is better)
- $R \geq 3$ rounds (5 recommended)
- ≥ 3 random seeds per task
- Agent must accept failure observations as input
- Trajectory recording capability (for $\hat{\Phi}_E$)

6 Discussion

Why not just report multi-trial success rate? Averaging over independent trials (the standard approach) captures the agent's expected performance—a measure of intelligence. WB-E's *sequential* trials with failure feedback capture the agent's ability to *improve*—a measure of wisdom. These are orthogonal quantities (as our $\rho = +0.09$ confirms). This distinction parallels the difference between IQ and experiential learning in cognitive psychology [11].

Relation to continual learning. WB-E is complementary to continual learning benchmarks such as CLEAR [5] and CTrL [12]: those measure resistance to catastrophic forgetting over training distribution shifts, while WB-E measures runtime adaptation within a single deployment. The Reflexion framework [7] demonstrated that verbal self-reflection improves multi-round performance; WB-E provides the standardized evaluation protocol that Reflexion lacked.

Implications for deployment. In long-term deployment (e.g., 24/7 warehouse operation [1]), agents encounter novel failure modes daily. An agent with high W_E continually improves; one with $W_E = 0$ stagnates. Over a year of deployment, the cumulative difference is massive—even if the high-W_E agent started with lower first-trial success.

Broader impact. WB-E enables more informed deployment decisions by revealing whether a robot can learn from operational failures. The risk is that high EWQ scores in simulation may not transfer to real-world settings, leading to overconfidence. We recommend WB-E scores be validated with real-world pilot studies before safety-critical deployment.

Limitations. (1) WB-E requires $5\times$ the evaluation compute of single-trial benchmarks. (2) The perturbation suite for $\hat{H}_E$ is simulator-dependent. (3) $R = 5$ rounds may not fully capture learning dynamics for very slow learners. (4) The axiom measurements depend on the quality of trajectory recording.

7 Conclusion

WisdomBench-Embodied provides the first standardized protocol for measuring learning ability in physical agents. By revealing the orthogonal dimension of wisdom alongside intelligence, WB-E enables the community to build robots that don't just perform—they *improve*. The benchmark is open-source and designed for immediate adoption atop any existing evaluation pipeline.

A call to action. We urge the community to report both I_E and EWQ in future work. A robot that succeeds on the first try is impressive. A robot that learns from failure is *wise*.

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